FINAL RESEARCH PROJECT - PROCESS REPORT

Next-Phase Research Execution Plan

A bounded path from READY screening to final research claims, with explicit storage, evidence, and submission controls.

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ACTIVE STAGE READY	RECOMMENDED NEXT SUBMISSIONS	CANONICAL GATES REMAINING

RECOMMENDED NEXT MOVE

Open a new screening window only after a fresh signed capacity pass. Use one screening-array submission, verify every task, then use one aggregation submission. Keep the window below five submissions. Full screening remains unsubmitted.

TECHNICAL SUMMARY

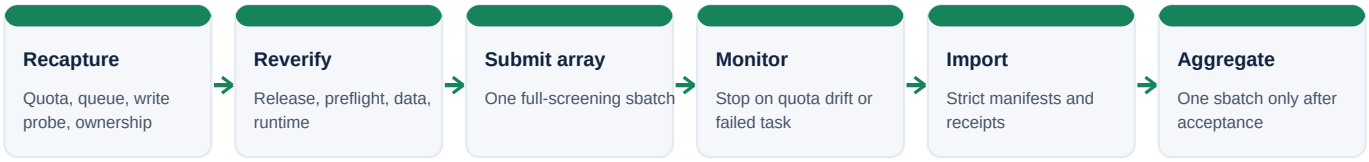
The project is ready for full screening in scientific dependency terms, but not under a standing operational authorization. The next window should recapture quota, verify the exact release and preflight, submit one array, validate all returned trial evidence, and submit aggregation only after the array is complete. Later phases remain dependency-gated so that compute scale cannot outrun research validity.

Campaign	large_scale_thai_drl_v2
Evidence cut	17 July 2026
Generated	20 July 2026
Audience	Research team, technical reviewer, and HPC operator

Evidence basis: active campaign state, storage-safe capacity policy, runtime projections from job 3550, and the preregistered 503-run campaign graph.

SECTION 1

Next controlled HPC window



PRE-SUBMISSION CHECKLIST

- Capture a fresh quota snapshot and decision no more than 15 minutes before submission.
- Confirm all eight group/project byte and inode checks pass, the 1 MiB write probe fsyncs and is removed, and projected margin is positive.
- Prefer more than the minimum margin. The last post-smoke pass had only about 0.309 GiB, so a group ownership or quota correction is the safer prerequisite.
- Confirm active release 56358f2a4996...6873b55e and preflight SHA 1ccac9827e2f...6ab59f0c.
- Confirm observed data identity, locked configuration identity, empty or nonconflicting queue state, and available compute credit.
- Render the exact command first and confirm it contains one sbatch invocation.

SUBMISSION ENVELOPE

Submission	When allowed	Purpose	Stop condition
1 - screening array	Fresh capacity and identity gates pass	Run 30 trials for each of five algorithms	Any quota, task, checksum, or checkpoint-path failure
2 - screening aggregate	All expected trial manifests and receipts pass strict import	Rank candidates using preregistered metrics	Missing trial, incompatible identity, or aggregate validation failure
3-4 - reserve only	Explicitly reopened window	Recovery or independent verification only	Never use merely to bypass a failed gate

ARRAY-COUNT CLARIFICATION

A Slurm array may create many task elements, but it is one reviewed sbatch submission. The project should still monitor every task independently and stop the campaign if the shared capacity contract is violated.

SECTION 2

Scientific campaign after screening

Phase	Main purpose	Acceptance evidence	Why it matters
Screening and aggregate	Identify competitive configurations across five algorithms	Complete trials, aggregate receipt, preregistered ranking	Prevents choosing a model from one lucky run
Training-budget sweep	Measure performance versus compute	Budget-specific trials and aggregate	Shows whether conclusions depend on spending more compute
Nested walk-forward	Separate tuning periods from evaluation periods	Fold-complete nested manifests and aggregate	Reduces optimistic selection bias
Protocol lock	Freeze decisions before final tests	Signed selected candidates and exact deployment	Prevents changing methods after seeing test outcomes
Robustness	Repeat across seeds, windows, and perturbations	Complete robustness matrix and aggregate	Tests stability rather than one-path success
Feature ablation	Remove data groups in controlled comparisons	Readiness-bound inputs and ablation aggregate	Measures which information truly contributes
Execution stress	Increase costs, slippage, and constraints	Stress manifests and validated comparison	Tests whether paper performance survives implementation friction
Finalize	Synthesize evidence and limitations	Final gate, tables, figures, archive	Separates accepted claims from exploratory findings

DATA ACQUISITION AND SCOPE

- Observed market OHLCV and market-context features are currently certified for execution.
- Observed macro and fundamental inputs must be imported, byte/hash certified, and included in a passing readiness receipt before related ablation claims.
- If those inputs cannot be obtained in time, amend the preregistered scope before accessing held-out test outcomes; do not silently replace missing observed data with synthetic evidence for the final claim.
- Generated or synthetic data may support pipeline tests and sensitivity analysis, but it must remain explicitly labeled and separated from observed-market evidence.

SECTION 3

Decision gates, risks, and completion criteria

Risk	Current evidence	Required mitigation	Owner decision
Group soft-quota margin	0.309 GiB after smoke; historical pass expired	Fresh pass plus more headroom or ownership/quota correction	HPC operator
Runtime extrapolation	All five projected below 72 h from one smoke trial each	Monitor first array elements; revise resource requests if observed rates drift	Research/HPC team
Partial or stale evidence	Strict imports and content-addressed identities implemented	Reject mismatched hashes, incomplete manifests, and unsafe reuse	Pipeline gate
Selection bias	Preregistration and dependency graph in place	Aggregate before selection; nested evaluation; protocol lock	Research lead
Missing external features	Macro/fundamental claims not readiness-certified	Certified observed inputs or preregistered scope amendment	Research lead
Overclaiming smoke results	Smoke proves execution only	Reserve performance claims for accepted aggregate and final gates	Report reviewer

PHASE ACCEPTANCE CHECKLIST

- Screening: every expected algorithm/trial completes with exact identities and zero prohibited checkpoint paths.
- Aggregation: selection follows preregistered metrics and reports uncertainty, failures, and missingness.
- Nested/robustness/ablation/stress: all dependency-specific release and data gates pass before compute is submitted.
- Finalization: conclusions distinguish exploratory, screening, and held-out evidence; limitations and resource costs are reported.
- Archive: configuration, code, data fingerprints, job accounting, logs, manifests, receipts, figures, and reports are content-addressed.

DEFINITION OF DONE

The project is complete only when the final campaign gate verifies the preregistered chain through screening, model selection, nested evaluation, robustness, ablation, execution stress, statistical synthesis, and reproducibility archive. More HPC capacity should accelerate this chain, not remove it.

NEXT-PHASE EVIDENCE INPUTS

Artifact or identifier	Verified value
Active campaign state	`campaigns/large_scale_thai_drl_v2/state.json`
Storage-safe capacity policy	`campaigns/large_scale_thai_drl_v2/operational/screening_capacity/README.md`
Locked screening configuration	`campaigns/large_scale_thai_drl_v2/derived/locked_large_scale_screening.yaml`
Verified smoke summary	`results/screening_storage_safe_smoke_3550/screening_smoke_summary.json`